



www.sciencemag.org/cgi/content/full/science.1192838/DC1

Supporting Online Material for

Cellodextrin Transport in Yeast for Improved Biofuel Production

Jonathan M. Galazka, Chaoguang Tian, William T. Beeson, Bruno Martinez,
N. Louise Glass, Jamie H. D. Cate*

*To whom correspondence should be addressed. E-mail: jcate@lbl.gov

Published 9 September 2010 on *Science Express*
DOI: 10.1126/science.1192838

This PDF file includes:

Materials and Methods

Figs S1 to S8

References

Supplementary Online MaterialTransporter and β -glucosidase orthologues

GenBank accession numbers or Joint Genome Institute (JGI) protein ID (PID) numbers for cellodextrin transporters are as follows: *Tuber melanosporum*, CAZ81962.1; *Pichia stipitis*, ABN65648.2; *Laccaria bicolor*, EDR07962, *Aspergillus oryzae*, BAE58341.1; *Phanerochaete chrysosporium*, PID 136620 (JGI) (S1); *Postia placenta*, PID 115604 (JGI) (S2). The GenBank accession number for *Saccharomyces cerevisiae* HXT1 and *Kluyveromyces lactis* LACP are DAA06789.1 and CAA30053.1, respectively. The *P. chrysosporium* and *P. placenta* genomes can be accessed at <http://genome.jgi-psf.org/Phchr1/Phchr1.home.html> and <http://genome.jgi-psf.org/Pospl1/Pospl1.home.html>, respectively.

GenBank accession numbers for cellodextrin hydrolases that are orthologues of NCU00130 (named *gh1-1*; in glycosyl hydrolase family GH1) are as follows: *T. melanosporum*, CAZ82985.1; *A. oryzae*, BAE57671.1; *P. placenta*, EED81359.1, and *P. chrysosporium*, BAE87009.1. The other organisms that contain cellodextrin transporter orthologues contain genes in the GH3 family predicted to be intracellular β -glucosidases (S3, S4), as follows: *Kluyveromyces lactis*, CAG99696.1; *Laccaria bicolor*, EDR09330; *Clavispora lusitaniae*, EEQ37997.1; and *Pichia stipitis*, ABN67130.1.

Materials and Methods

Strains

The yeast strain used in this study was YPH499 (S5), which has the genotype: *MATa ura3-52 lys2-801_amber ade2-101_ochre trp1-Δ63 his3-Δ200 leu2-Δ1*. It was grown in YPD media supplemented to 100 mg/L adenine hemisulfate. Transformed strains (S6) were grown in the appropriate complete minimal dropout media, supplemented to 100 mg/L adenine hemisulfate. *Neurospora crassa* strains used in this study were obtained from the Fungal Genetics Stock Center (FGSC) (S7) and include WT (FGSC 2489) and two cellobiose transporter deletion strains (FGSC 16575, ΔNCU00801.2 and FGSC 17868, ΔNCU08114.2 (S8)).

Plasmids and Cloning

Transporters were cloned into the 2μ plasmid, pRS426, which was modified to include the *S. cerevisiae* PGK1 promoter inserted between SacI and SpeI using the primers, ATATATGAGCTCGTGAGTAAGGAAAGAGTGAGGAACTATC and ATATATACTAGTTGTTTTATATTTGTTGTAAAAAGTAGATAATTACTTCC (In all primers above and below, restriction sites are underlined). NCU00801 with a C-terminal Myc-tag and optimized Kozak sequence (S9) was then inserted between BamHI and EcoRI using the primers,

ATGGATCCAAAAATGTCGTCTCACGGCTCC and
 ATGAATTCCTACAAATCTTCTTCAGAAATCAATTTTTGTTTCAGCAACGATAGC
 TTCGGAC, and NCU08114 with a C-terminal Myc-tag and optimized Kozak
 sequence was inserted between *SpeI* and *Clal* using the primers,
 ATACTAGTAAAAATGGGCATCTTCAACAAGAAGC and
 GCATATCGATCTACAAATCTTCTTCAGAAATCAATTTTTGTTTCAGCAACAGAC
 TTGCCCTCATG. To make GFP fusions, superfolder GFP (*S10*) with an N-
 terminal linker of Gly-Ser-Gly-Ser was first inserted between the *Clal* and *Sall*
 sites of the PGK1 promoter-containing pRS426 plasmid with the primers,
 TATTAAATCGATGGTAGTGGTAGTGTGAGCAAGGGCGAGGAG and
 TATTAAGTCGACCTACTTGTACAGCTCGTCCATGCC. Transporters were then
 fused to GFP as follows: NCU00801 was inserted between *BamHI* and *EcoRI*
 using the primers, GCATGGATCCATGTCGTCTCACGGCTCC and
 TATAATGAATTCAGCAACGATAGCTTCGGAC, and NCU08114 was inserted
 between *SpeI* and *EcoRI* using the primers,
 TATTAAACTAGTATGGGCATCTTCAACAAGAAGC and
 TTATAAGAATTCAGCAACAGACTTGCCCTCATG.

The β -glucosidase, NCU00130, was cloned into the 2 μ plasmid, pRS425,
 modified to include the PGK1 promoter described above. NCU00130 with an
 optimized Kozak sequence and a C-terminal 6xHis tag was inserted between
SpeI and *PstI* using the primers,
 GCATACTAGTAAAAATGTCTCTTCCTAAGGATTCCTCT and
 ATACTGCAGTTAAATGATGATGATGATGATGGTCCTTCTTGATCAAAGAGTCA

AAG. All constructs included the Cyc transcriptional terminator between XhoI and KpnI. All *N. crassa* genes were amplified by PCR from cDNA synthesized from mRNA isolated from *N. crassa* (FGSC 2489) cultured on minimal media with pure cellulose (Avicel) as the sole carbon source.

Yeast Growth Assays

To monitor growth on cellodextrins, engineered strains were grown in 5 mL of complete minimal media with appropriate dropouts, overnight. These starter cultures were washed 3x with 25 mL of ddH₂O, and resuspended to an O.D. (600 nm) of 0.1 in Yeast Nitrogen Base (YNB) plus the appropriate Complete Supplemental Media (CSM) and 1% (w/v) cellobiose, or 0.5% (w/v) of either cellotriose, or cellotetraose. Assays were performed in a Bioscreen C™ with constant shaking at maximum amplitude at 30 °C, and a final assay volume of 0.4 mL. The change in O.D. was measured either at 600 nm or using a wideband filter from 450 – 580 nm. Growth rates were taken from the linear portion of each growth curve, and are reported as the mean of three independent experiments ± the standard deviation between these experiments. Cellotriose and cellotetraose was obtained from Seikagaku Biobusiness Corporation.

Purification of NCU00130 (named GH1-1) and Assay of its activity

A 1 L culture of *S. cerevisiae* expressing *cdt-1* and *gh1-1* was grown to an O.D. (600 nm) of 2.0 in complete minimal media. Cells were harvested by centrifugation, and resuspended in 30 mL of lysis buffer (50 mM NaH₂PO₄ [pH

8.0], 300 mM NaCl, 10 mM imidazole, 2 mM β -ME, Complete™ Mini, EDTA free protease inhibitor cocktail). Cells were lysed by sonication, and the lysate cleared by centrifugation at 15,000 g for 30 minutes. The lysate was bound to 1 mL of Ni-NTA resin by gravity flow, and washed 3x with 25 mL wash buffer (Identical to lysis buffer but with 20 mM imidazole). GH1-1 was eluted with 5 mL of elution buffer (Identical to lysis buffer but with 250 mM imidazole), and the appropriate fractions were pooled, exchanged into storage buffer (Phosphate Buffered Saline, 2 mM DTT, 10% glycerol), aliquoted, frozen in liquid nitrogen, and stored at -80 °C. Purity was determined by SDS-PAGE (Fig. S9), and protein concentration determined from the absorbance at 280 nm, using an extinction coefficient of 108750 M⁻¹ cm⁻¹.

Purified GH1-1 was assayed for hydrolysis activity with different cellodextrin substrates. Activity was measured by incubating 5 pmol of enzyme with 500 μ M of each sugar in 150 μ L Phosphate Buffered Saline (PBS) plus 3 mM DTT. Reactions proceeded for 40 minutes at 30 °C, before 100 μ L was removed and quenched in 400 μ L of 0.1 M NaOH. The results were analyzed by ion chromatography with a Dionex ICS-3000, using a CarboPac PA200 column. Peaks were detected with an electrochemical detector.

Phylogenetic Analysis of Transporter Orthologues

Amino acid sequences of orthologues of *cdt-1* and *cdt-2* were obtained from online databases. Multiple sequence alignments were performed using T-

Coffee (S11). A maximum likelihood phylogeny was determined using PhyML version 3.0 (S12) with 100 bootstraps. Both programs were accessed through Phylogeny.fr (<http://www.phylogeny.fr/>). The resulting tree was visualized with FigTree v.1.2.1 (<http://tree.bio.ed.ac.uk/>).

Fermentation and SSF

In fermentation and SSF experiments, comparisons were made between yeast expressing *gh1-1* and either Myc-tagged *cdt-1*, or no transporter. These strains were grown aerobically overnight in complete minimal media, washed 3x with 25 mL water, and resuspended to a final O.D. of 2.0 in 50 mL YNB plus the appropriate CSM, and either 2% (w/v) cellobiose or 3% (w/v) pure cellulose (Avicel), in sealed serum flasks. The SSF reactions also included 50 Filter Paper Units/g cellulose of filter-sterilized Celluclast (Sigma C2730), without β -glucosidase supplementation. Reactions were carried out anaerobically at 30 °C with shaking. At indicated time points, 1 mL samples were removed and filtered through a 0.2 μ m syringe filter. The ethanol, glucose, and cellobiose concentration in the filtrate was determined by HPLC with an Aminex 87H column and refractive index detection.

N. crassa Growth and Alamar Blue[®] Assays

WT *N. crassa* (FGSC 2489), and the homokaryotic NCU08114 (FGSC 17868) deletion strain (S8) were acquired from the FGSC (S7), and grown at 25 °C in 50 mL of Vogel's salts plus 2% (w/v) of either sucrose or pure cellulose

(Avicel) in a 250 mL unbaffled flask using an inoculum of 10^6 conidia/mL. After 16 or 28 hours, respectively, 100 μ L of Alamar Blue[®] was added, and cultures incubated at room temperature for 20 minutes. At this time, 1 mL samples were removed, debris pelleted, and the fluorescence of 100 μ L of the supernatant determined with excitation/emission wavelengths of 535/595 nm in a Beckman Coulter Paradigm plate reader.

N. crassa Cellobiose Transport Assays

WT *N. crassa* (FGSC 2489), and homokaryotic deletion lines (S8) of NCU00801 (FGSC 16575) and NCU08114 (FGSC 17868) were acquired from the FGSC (S7), and grown for 16 hours in 50 mL Vogel's salts plus 2% (w/v) sucrose at 25 °C, starting with an inoculum of 10^6 conidia/mL. Mycelia were harvested by centrifugation, washed 3x with Vogel's salts, and transferred to Vogel's salts plus 0.5% (w/v) pure cellulose (Avicel) for 4 hours to induce the transporter expression. 10 mL of the culture was harvested by centrifugation, washed 3x with Vogel's salts, and resuspended in 1 mL ddH₂O plus cycloheximide (100 μ g/mL) and 90 μ M of the respective cellodextrin. To measure cellodextrin consumption, 100 μ L was removed after 15 minutes, clarified by centrifugation, and transferred into 900 μ L of 0.1 M NaOH. The amount of sugar remaining in the supernatant was determined by HPLC with a Dionex ICS-3000, using a CarboPac PA200 column. Peaks were detected with an electrochemical detector.

GFP Fluorescence and Confocal Fluorescence Microscopy

Bulk-cell GFP fluorescence measurements were made in a Beckman Coulter Paradigm plate reader with excitation/emission wavelengths of 485/535 nm. Confocal Fluorescence Microscopy was performed with cells at an O.D. (600 nm) of 0.8 – 1.2, using a 100x 1.4 NA oil immersion objective on a Leica SD6000 microscope attached to a Yokogawa CSU-X1 spinning disc head with a 488 nm laser and controlled by Metamorph software. Z series were recorded with a 200 nm step size and analyzed using ImageJ.

[³H] Cellobiose Transport Assays and Kinetic Parameters

Transport assays were performed using a modification of the oil-stop method (S13). Yeast strains expressing either *cdt-1* or *cdt-2* fused to GFP were grown to an O.D. (600 nm) of 1.5 – 3.0 in selective media, washed 3x with ice cold assay buffer (30 mM MES-NaOH [pH 5.6] and 50 mM ethanol), and resuspended to an O.D. of 20. To start transport reactions, 50 μ L of cells were added to 50 μ L of [³H]-cellobiose layered over 100 μ L of silicone oil (Sigma 85419). Reactions were stopped by spinning cells through oil for 1 minute at 17,000 g, tubes were frozen in ethanol/dry ice, and tube-bottoms containing the cell-pellets were clipped off into 1 mL of 0.5 M NaOH. The pellets were solubilized overnight, 5 mL of Ultima Gold scintillation fluid added, and CPM determined in a Tri-Carb 2900TR scintillation counter. [³H]-cellobiose was purchased from Moravsek Biochemicals, Inc., and had a specific activity of 4 Ci/mmol and a purity of >99%. Kinetic parameters were determined by

measuring the linear rate of [^3H]-cellobiose uptake over 3 minutes for cellobiose concentrations between 0.5 and 200 μM . V_{max} and K_{M} values were determined by fitting a single rectangular, 2-parameter hyperbolic function to a plot of rates vs. cellobiose concentration by non-linear regression in SigmaPlot[®]. V_{max} values were normalized for differences in transporter abundance by measuring the GFP fluorescence from 100 μL of cells at O.D. 20 immediately before beginning transport assays. Kinetic parameters reported in the text are the mean $\pm$ SEM from three separate experiments. Competition assays were performed by measuring transport of 50 μM [^3H]-cellobiose over 20 s in the presence of 250 μM of the respective competitors.

Measurement of β -glucosidase activity in culture media

To assay for contaminating β -glucosidase activity, strains were grown in 5 mL of complete minimal media with appropriate dropouts, overnight. For large scale experiments, these starter cultures were washed 3x with 25 mL ddH₂O and resuspended to an O.D. (600 nm) of 0.1 in 50 mL of Yeast Nitrogen Base (YNB) plus the appropriate Complete Supplemental Media (CSM) and 2% (w/v) cellobiose in 250 mL shaker flasks. Cultures were grown at 30 °C for 84 hours, at which point the O.D. (600 nm) was determined, and the cell-free culture supernatant collected by filtration through a 0.2 μm filter. For small scale experiments, starter cultures were washed as above, and resuspended to an O.D. (600 nm) of 0.1 in 400 μL of YNB plus the appropriate CSM and 0.5% (w/v) of either cellotriose or cellotetraose. Cultures were grown at 30 °C for 96 hours in

a Bioscreen C™ at which point the O.D. (600 nm) was determined, and the cell-free culture supernatant collected as above.

Filtrates were colorimetrically assayed for β -glucosidase activity by monitoring the release of *para*-Nitrophenol (pNP) from *para*-Nitrophenyl β -D-glucopyranoside (pNPG) (S16). Briefly, 80 μ L of the filtrate was mixed with 20 μ L of 5 mM pNPG and incubated for 1 hour at 30 °C. At this point, 133 μ L of 0.4 M glycine-NaOH [pH 10.8] was added, and the absorbance at 405 nm determined in a Beckman Coulter Paradigm plate reader. Purified NCU00130 (GH1-1, Fig. S5) was used as a control.

Large Scale Yeast Growth

To monitor growth on different carbon sources, engineered strains were grown in 5 mL of complete minimal media with appropriate dropouts, overnight. These starter cultures were washed 3x with 25 mL of ddH₂O, and resuspended to an O.D. (600 nm) of 0.1 in 50 mL YNB plus the appropriate CSM and 2% (w/v) cellobiose. Cultures were grown in 250 mL unbaffled flasks at 30 °C, with shaking at 200 RPM. The change in O.D. (600 nm) was monitored by periodically removing samples.

Supplementary Figures

Figure S1

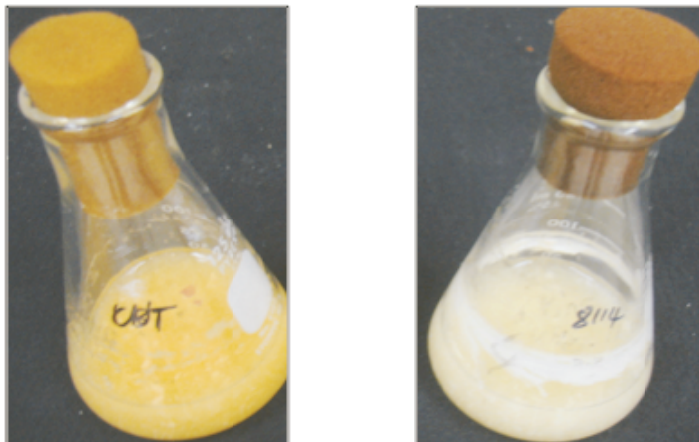
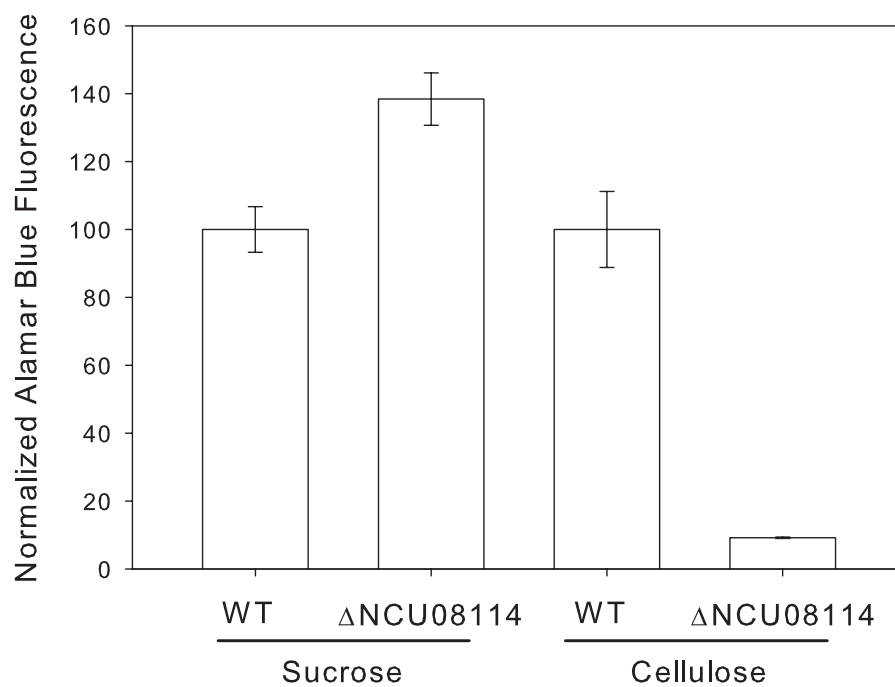
A**B**

Fig. S1. Growth phenotype of a *N. crassa* strain lacking NCU08114. **(A)** Shaker flasks of WT (left) and Δ NCU08114 (right) *N. crassa* strains after 3 days of growth with crystalline cellulose as a carbon source. **(B)** Alamar Blue[®] fluorescence from *N. crassa* cultures grown with either sucrose or crystalline

cellulose as a carbon source for 16 or 28 hours, respectively (average of three biological replicates $\pm$ SD). Fluorescence was normalized by setting WT to 100%. *N. crassa* lacking NCU00801 did not have an obvious phenotype (not shown).

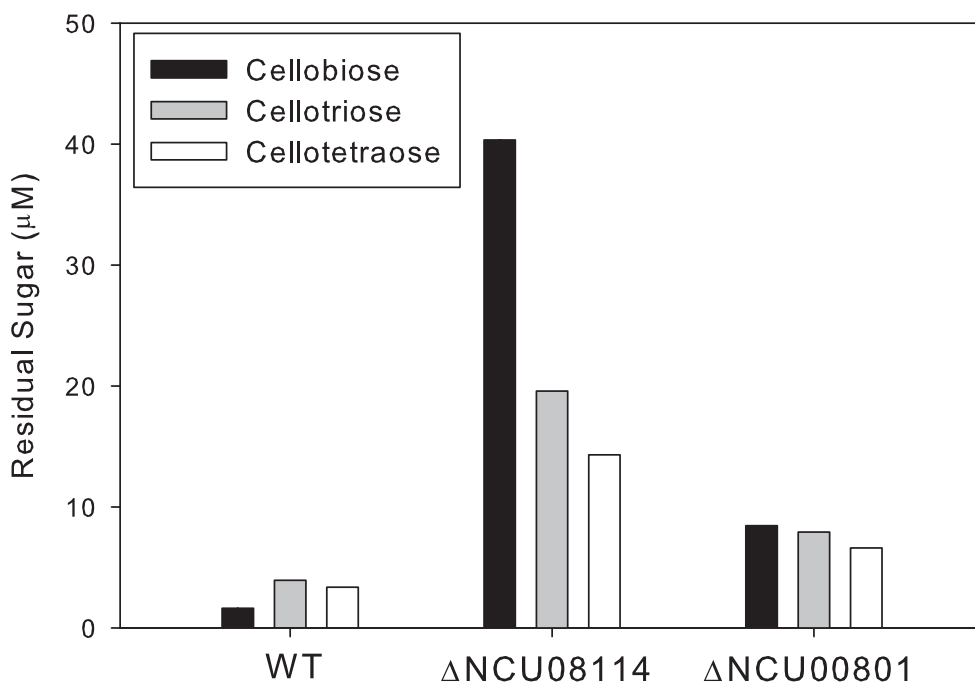
Figure S2

Fig. S2. Cellodextrin consumption by *N. crassa* strains lacking NCU00801 or NCU08114. The indicated *N. crassa* strains were incubated with 90 μM of the respective sugars for 15 minutes. Bars represent the mean concentration of sugars remaining in the supernatant following the incubation from two independent experiments. *N. crassa* secretes β-glucosidases (S14) that hydrolyze cellodextrins to glucose, which is subsequently taken up by monosaccharide transporters (S15). This alternate route of consumption leads to an underestimate of the cellodextrin transport defect in these deletion lines.

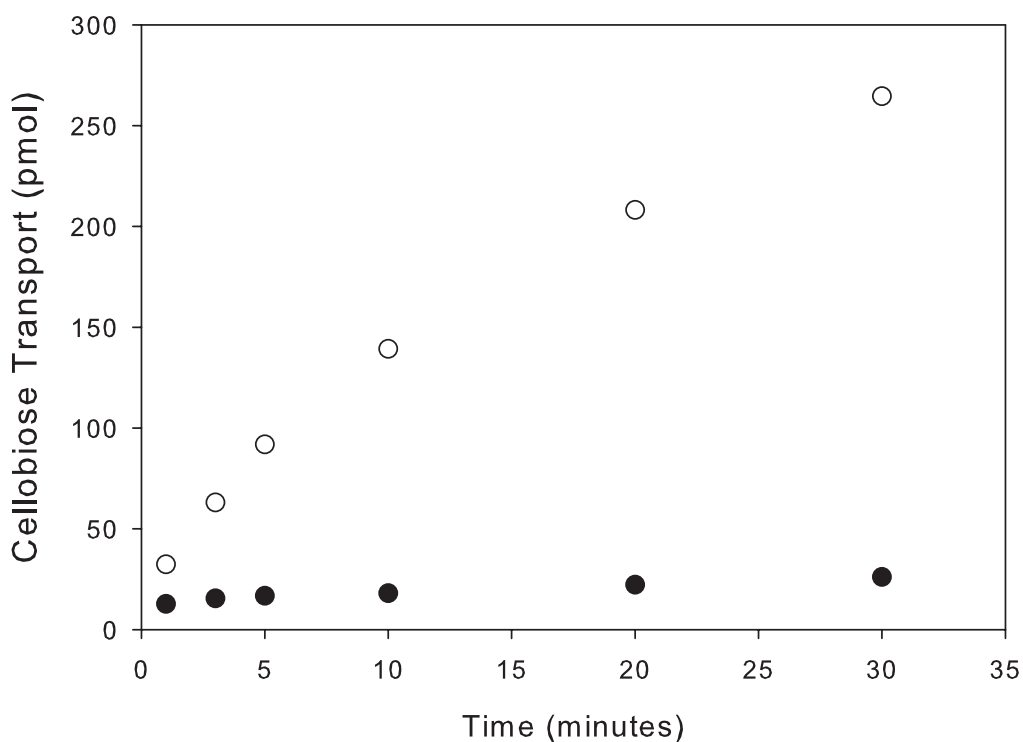
Figure S3

Fig. S3. Cellobiose transport by a *S. cerevisiae* strain expressing *cdt-1*. Shown is cellobiose transport by yeast with (○) or without (●) CDT-1. Both strains express the intracellular β -glucosidase, NCU00130 (named GH1-1). The initial concentration of cellobiose was 50 μ M. All values are the mean between two measurements.

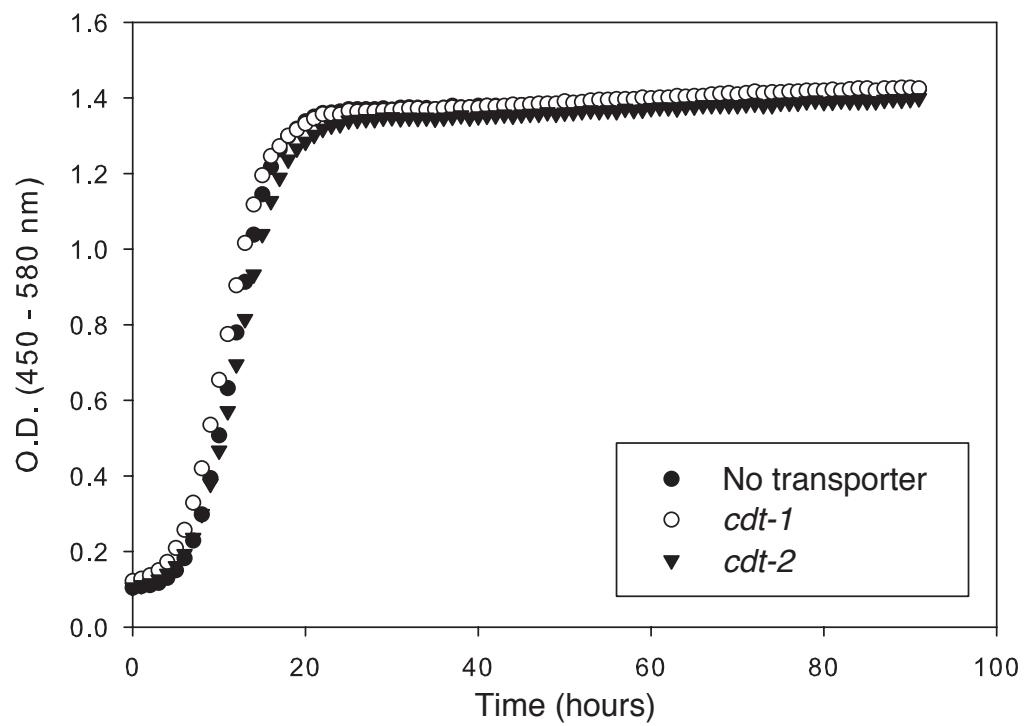
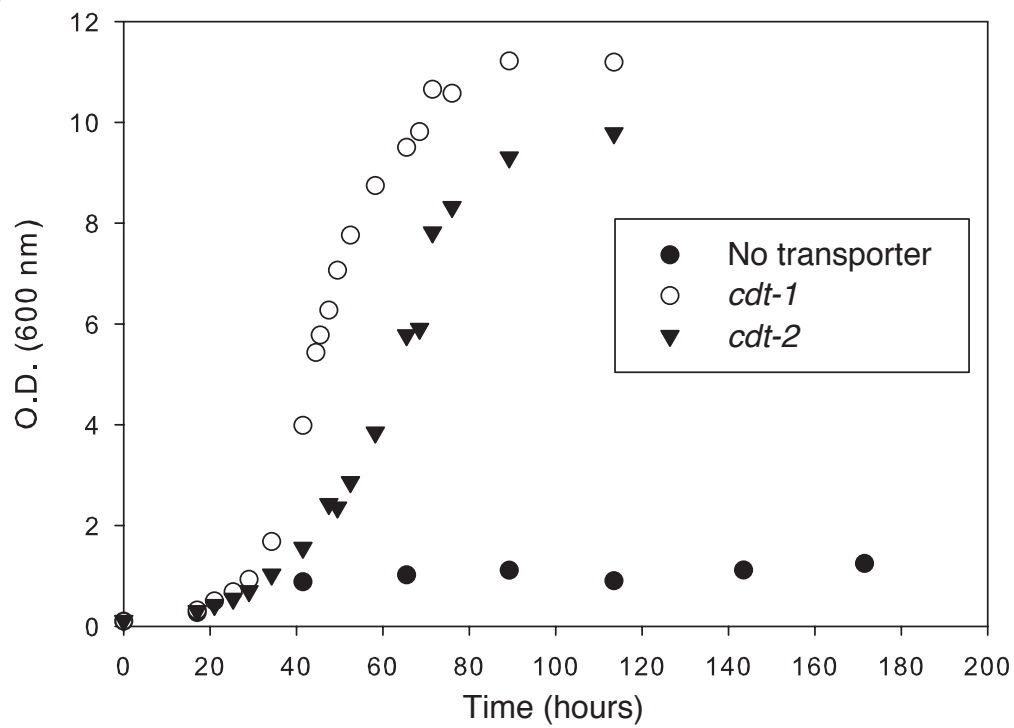
Figure S4**A****B**

Fig. S4. Growth of *S. cerevisiae* strains on glucose and cellobiose. **(A)** Growth of the strains on glucose using the Bioscreen-C™, for comparison to Figure 1B, 1C. Shown is the growth of yeast strains expressing *cdt-1* (○), *cdt-2* (▼), or no transporter (●). All strains express the β -glucosidase *gh1-1*. A representative experiment is shown. Growth rates from three independent growth experiments are as follows: *cdt-1*, $0.110 \pm 0.014 \text{ hr}^{-1}$; *cdt-2*, $0.116 \pm 0.003 \text{ hr}^{-1}$; no transporter, $0.131 \pm 0.0002 \text{ hr}^{-1}$ (mean $\pm$ SD). **(B)** Growth of the strains on cellobiose in 250 mL flasks. Values represent the mean O.D. between two replicate cultures of yeast strains expressing the β -glucosidase *gh1-1*, *cdt-1* or *cdt-2*, or a strain expressing *gh1-1*, but lacking any transporters. Approximate doubling times of yeast in these experiments is 3.2 hours (*cdt-1*) and 4.7 hours (*cdt-2*).

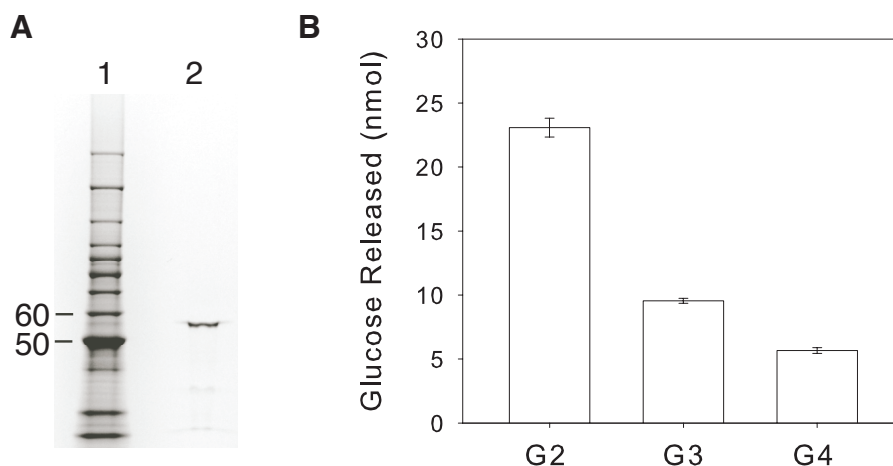
Figure S5

Fig. S5. Hydrolysis of cellooligosaccharides by NCU00130 (GH1-1). **(A)** SDS-PAGE gel of purified GH1-1. Lane 1, Protein molecular weight standards, in kDa. Lane 2, GH1-1 after purification over nickel-NTA resin. Molecular weights in kDa are shown to the left. **(B)** Glucose produced from cellobiose (G2), cellotriose (G3), and cellotetraose (G4) hydrolysis by purified GH1-1 (average of three independent experiments $\pm$ SD). Residual glucose in incubations without enzyme (2 nmol) was subtracted from the values shown.

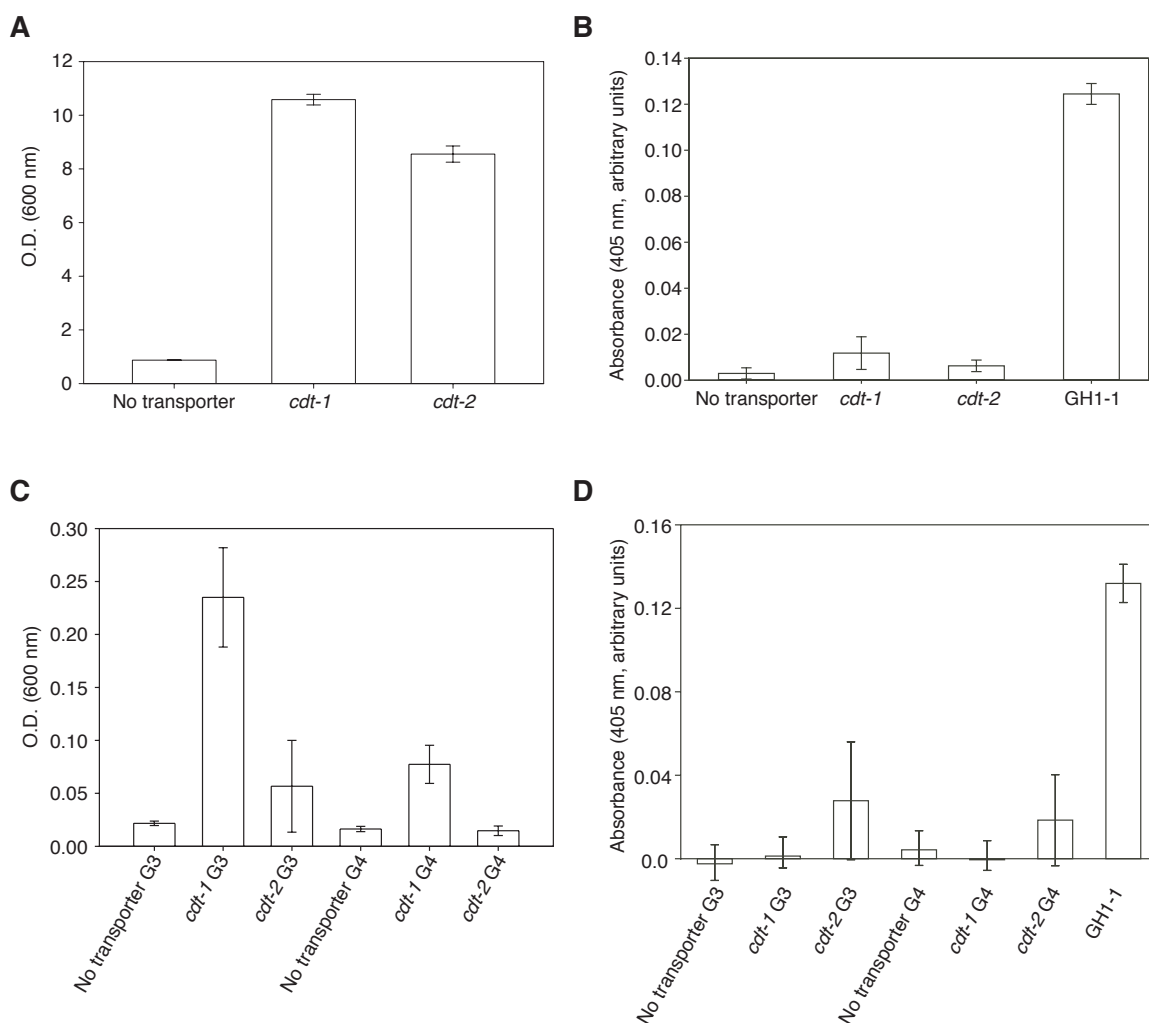
Figure S6

Fig. S6. β -glucosidase activity in cell culture broth after growth experiments. **(A)** Final O.D. of yeast cultures grown in 250 mL shake flasks for 84 hours (mean of three technical replicates $\pm$ SD) with cellobiose as a sole carbon source. **(B)** β -glucosidase activity measured by *p*-Nitrophenol generation (mean of three technical replicates $\pm$ SD). Filtered culture broths from the cultures in (A) were used. Purified β -glucosidase GH1-1 (6.3 nM) was used as a positive control. **(C)** Final O.D. of yeast cultures grown in a Bioscreen C™ for 96 hours (mean of three biological replicates $\pm$ SD) with cellotriose or cellotetraose as a sole carbon

source. **(D)** β -glucosidase activity measured by *p*-Nitrophenol generation (mean of three biological replicates $\pm$ SD). Filtered culture broths from the cultures in (C) were used. Purified β -glucosidase GH1-1 (6.3 nM) was used as a positive control.

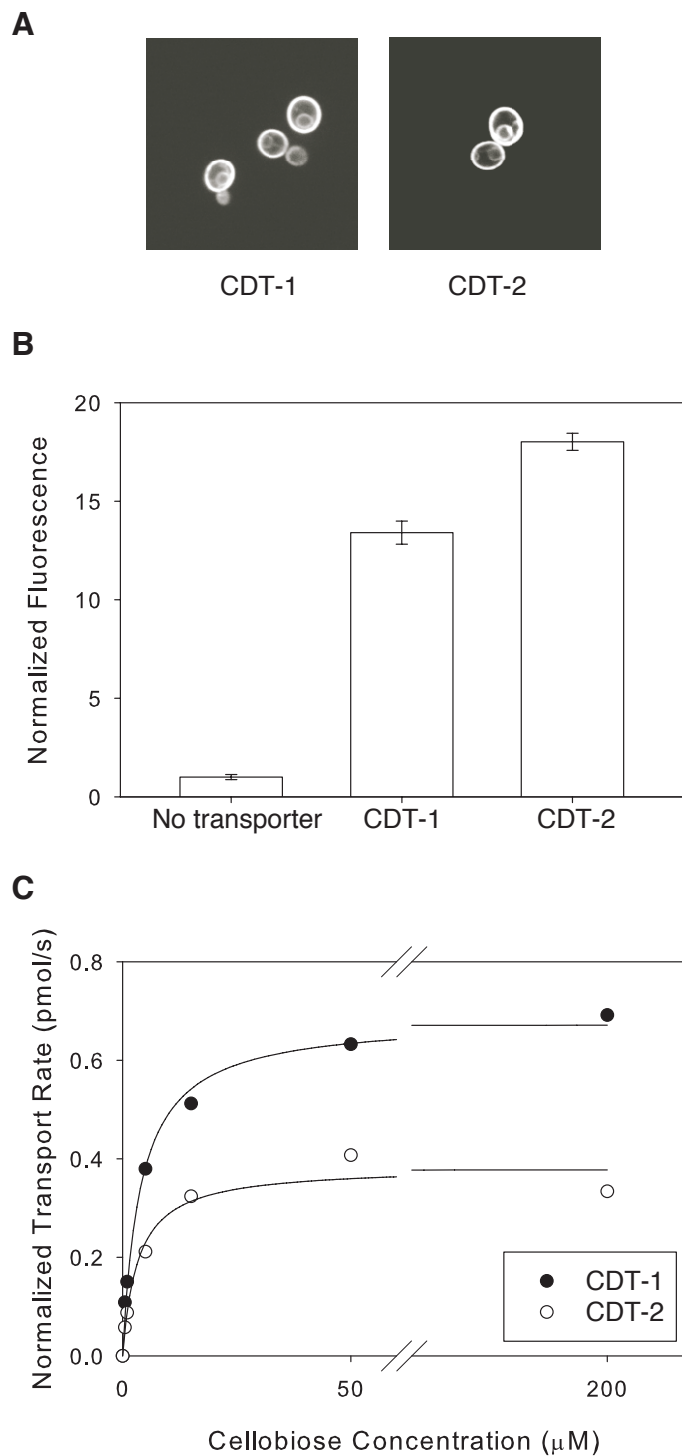
Figure S7

Fig. S7. Kinetics of cellobiose transport by CDT-1 and CDT-2. **(A)** Localization of GFP fused to CDT-1 and CDT-2. Images of *S. cerevisiae* strains expressing *cdt*-

1 (left), or *cdt-2* (right), fused to GFP at their C-terminus. **(B)** GFP fluorescence of yeast strains without a cellobiose transporter, or expressing *cdt-1* or *cdt-2* fused to GFP at their C-terminus (average of three biological replicates $\pm$ SD).

(C) The rate of cellobiose transport was determined as a function of cellobiose concentration by yeast strains expressing either *cdt-1* or *cdt-2*. The transport rate was normalized for transporter abundance.

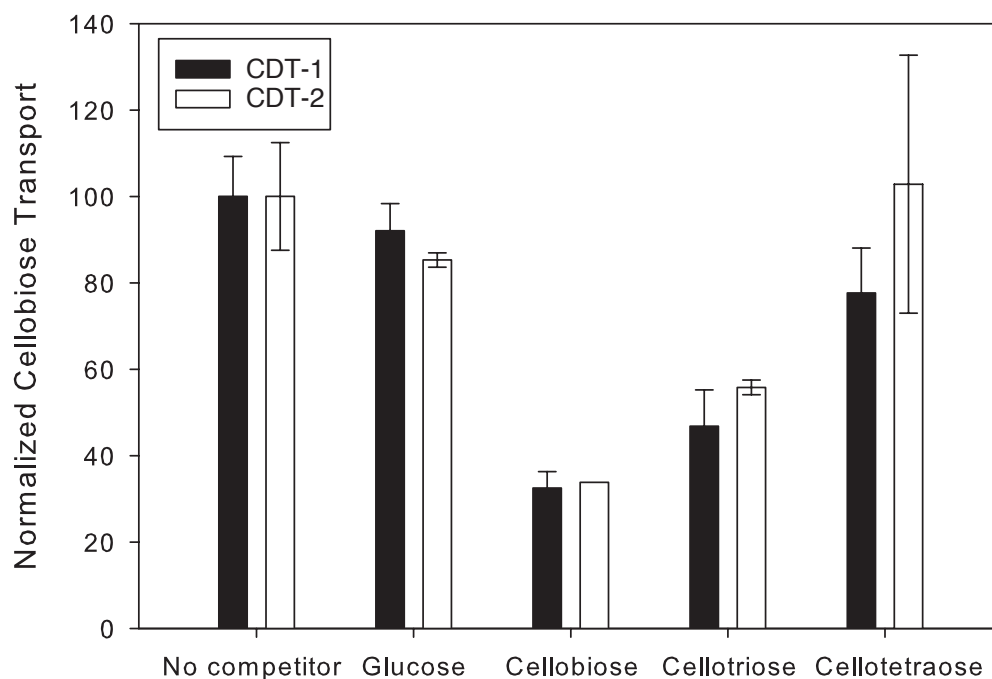
Figure S8

Fig. S8. Competition by cellodextrins for cellobiose transport in strains expressing *cdt-1* or *cdt-2*. A 5-fold excess of the respective unlabeled sugar was included during assays of [^3H]-cellobiose transport. Substrates of CDT-1 or CDT-2 should decrease the [^3H]-cellobiose transport rate by competing for binding. Values (average of three independent experiments $\pm$ SD) were normalized by setting the rate of [^3H]-cellobiose transport without a competing sugar to 100.

References

- S1. D. Martinez *et al.*, *Nature Biotechnology* **22**, 695 (2004).
- S2. D. Martinez *et al.*, *Proc Natl Acad Sci U S A* **106**, 1954 (2009).
- S3. J. D. Bendtsen, H. Nielsen, G. von Heijne, S. Brunak, *J Mol Biol* **340**, 783 (Jul 16, 2004).
- S4. B. L. Cantarel *et al.*, *Nucleic Acids Res* **37**, D233 (Jan, 2009).
- S5. R. S. Sikorski, P. Hieter, *Genetics* **122**, 19 (May, 1989).
- S6. D. M. Becker, V. Lundblad, *Curr Protoc Mol Biol* **Chapter 13**, Unit13 7 (May, 2001).
- S7. K. McCluskey, *Adv Appl Microbiol* **52**, 245 (2003).
- S8. H. V. Colot *et al.*, *Proc Natl Acad Sci U S A* **103**, 10352 (Jul 5, 2006).
- S9. H. Miyasaka, *Yeast* **15**, 633 (Jun 15, 1999).
- S10. J. D. Pedelacq, S. Cabantous, T. Tran, T. C. Terwilliger, G. S. Waldo, *Nat Biotechnol* **24**, 79 (Jan, 2006).
- S11. C. Notredame, D. G. Higgins, J. Heringa, *J Mol Biol* **302**, 205 (Sep 8, 2000).
- S12. S. Guindon, O. Gascuel, *Syst Biol* **52**, 696 (Oct, 2003).
- S13. C. S. Arendt, K. Ri, P. A. Yates, B. Ullman, *Anal Biochem* **365**, 185 (Jun 15, 2007).
- S14. C. Tian *et al.*, *Proc Natl Acad Sci U S A* **106**, 22157 (Dec 29, 2009).
- S15. G. A. Scarborough, *Int Rev Cytol* **34**, 103 (1973).
- S16. T. M. Wood, K. M. Bhat, *Methods in Enzymology* **160**, 87-112 (1988).